INTELLIGENT SYSTEM FINAL PROJECT

Chess Engine with CNN & Search

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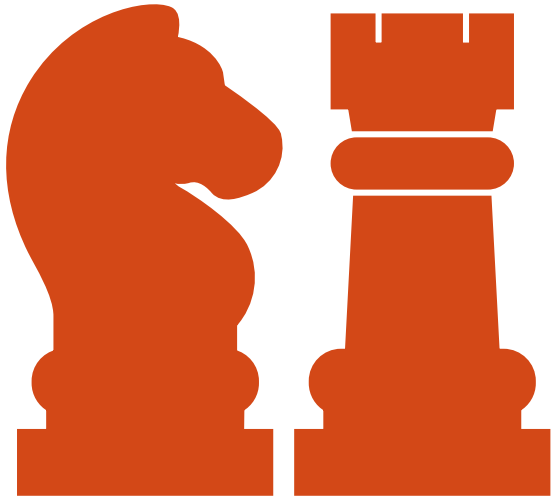
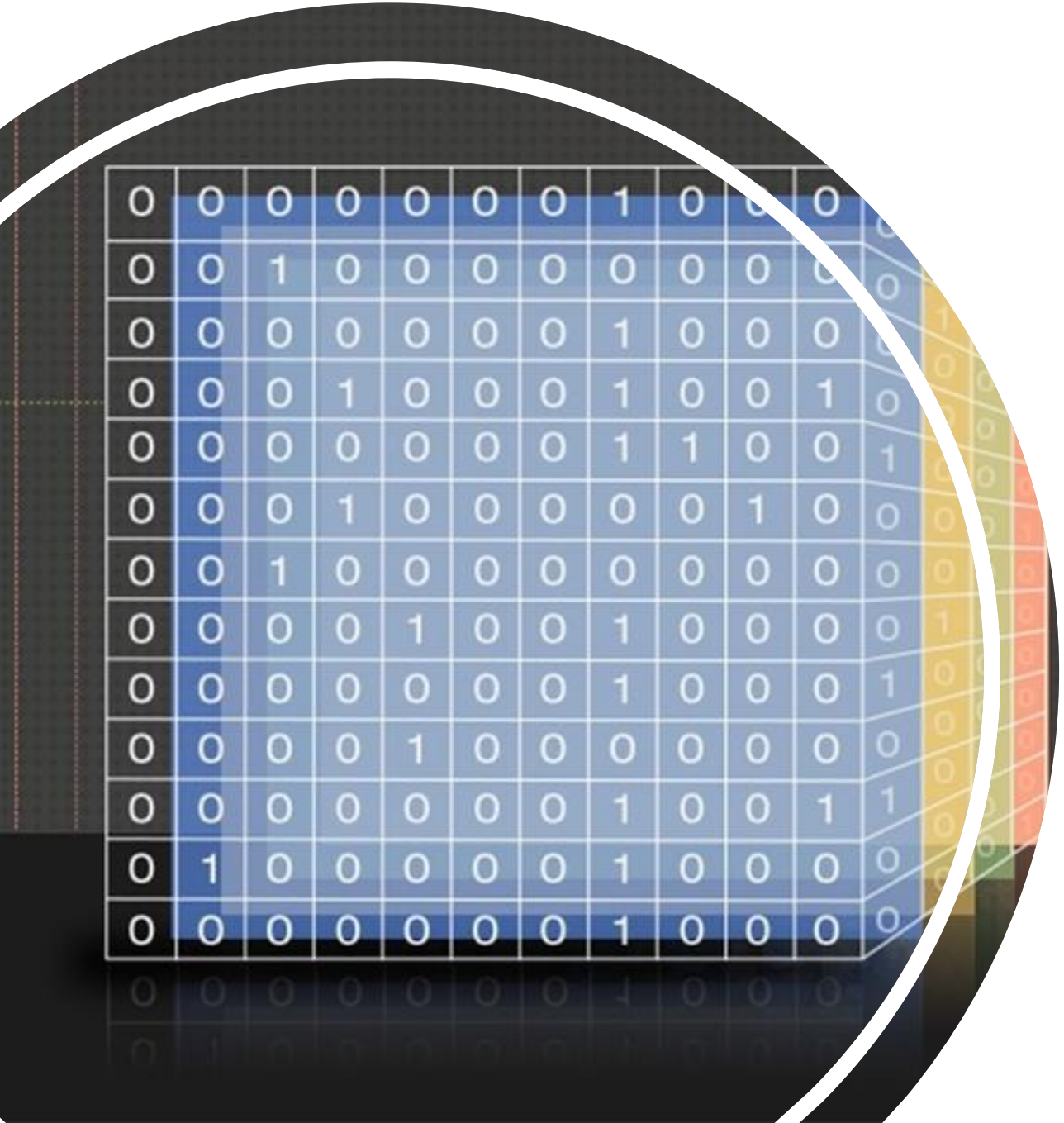
PROBLEM DEFINITION

- Goal: Predict evaluation score of a chess position
- Regression problem (continuous output)
- Use evaluation to guide move selection
- Inspired by engines like Stockfish / AlphaZero



DATA COLLECTION

- Dataset: Chess Evaluations
  - 16 million chess positions with Stockfish evaluation at a depth of 22
- Each sample:
  - FEN string (board state)
  - Evaluation score (centipawns)
- Used subset of dataset for training

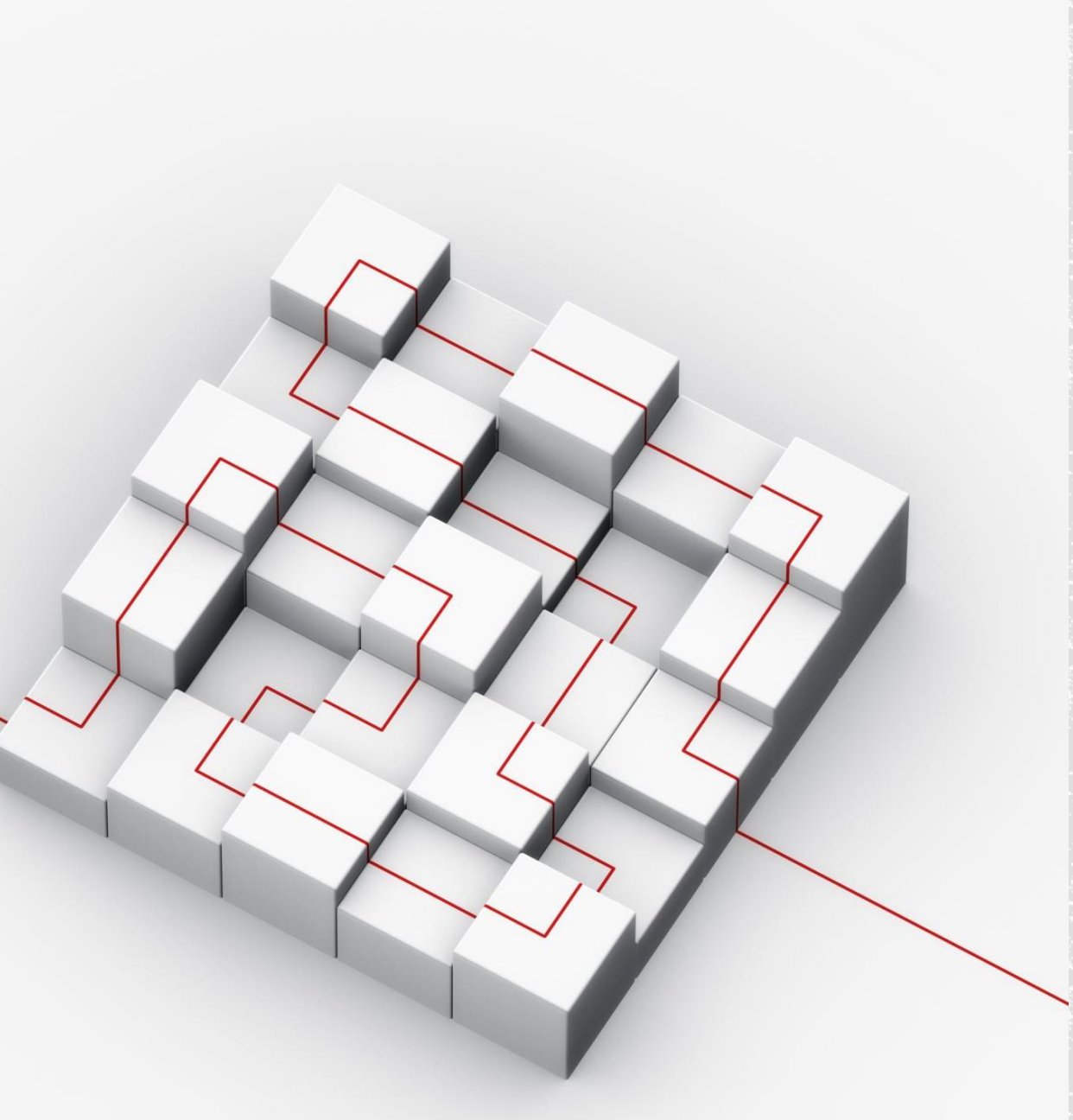


DATA PREPARATION

- Converted FEN → tensor representation
- Tensor shape: **19 × 8 × 8**
- Includes:
  - 12 planes for pieces
  - Side to move
  - Castling rights
  - En passant square
  - Half-move clock

EVALUATION SCALING

- Raw evaluations can be very large
- Applied normalization:
  - tanh(evaluation / 400)
- Benefits:
  - Stabilizes training
  - Keeps values in range [-1, 1]

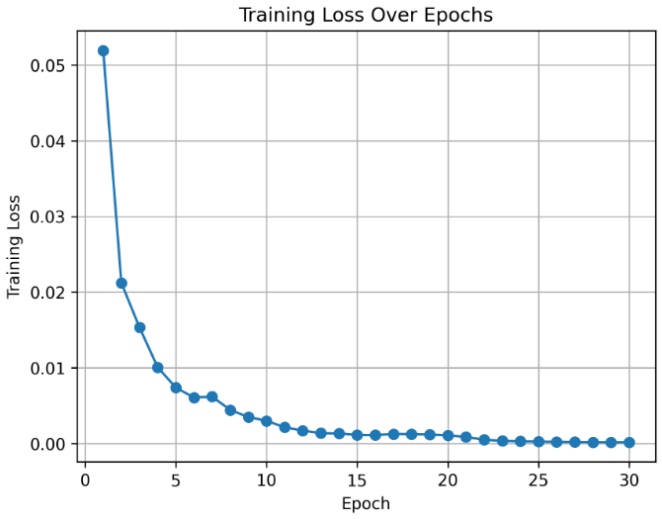


CNN ARCHITECTURE

- Input 19-channel board tensor
- Convolutional layer → feature extraction
- 20 Residual Blocks → deeper learning
- Value head → scalar output
- **Why CNN?**
  - Captures spatial relationships
  - Learns board patterns

TRAINING PROCESS

- Loss functions:
  - MSE
  - Smooth L1
- Optimizers:
  - Adam/AdamW
- Learning rate scheduler used
- Train/test split applied **(70/30)**

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Epoch 30/30
Train Loss: 0.0002
Test  Loss: 0.0070
MAE: 0.0724 | RMSE: 0.1186 | R2: 0.8955
----------------------------------------
Average Training Loss: 0.0049
Average Testing Loss: 0.0105
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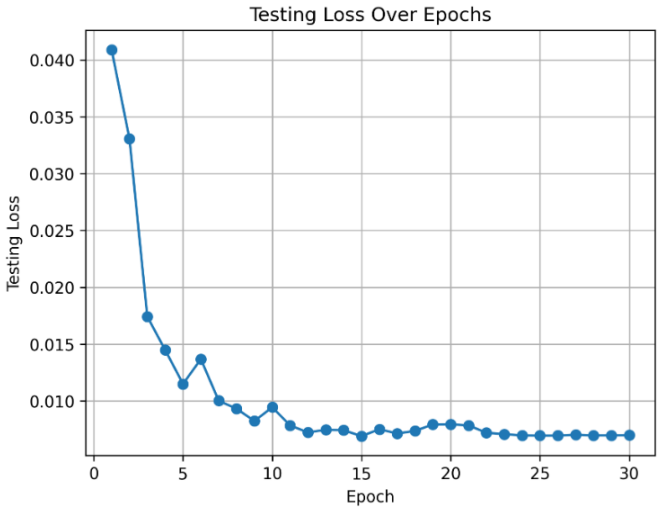


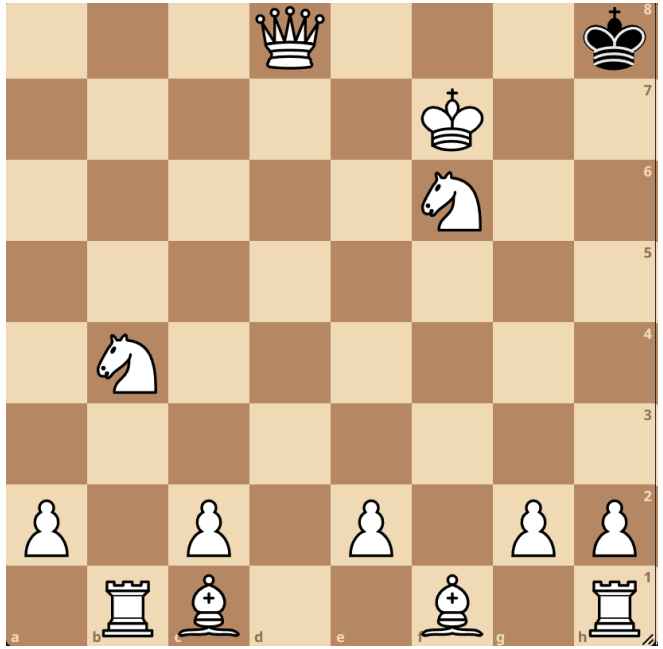
RESULTS

- Observations
  - Our model performed best with a AdamW optimizer and SmoothL1 loss function

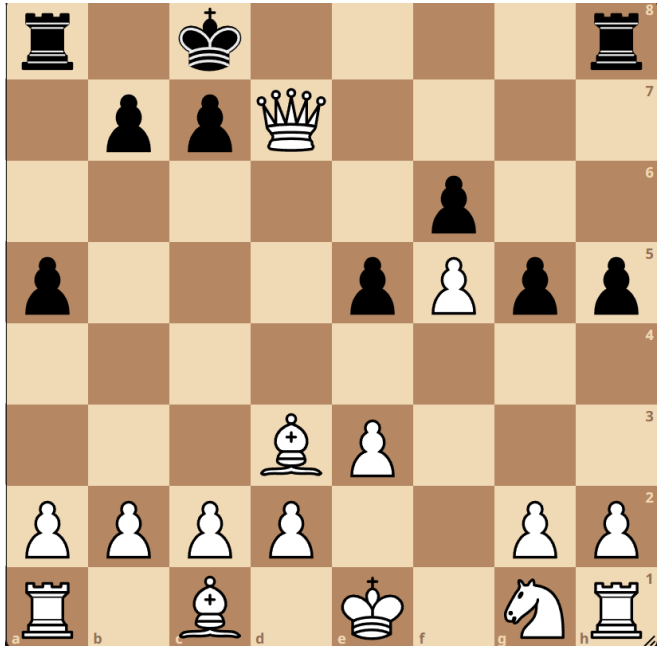
WHY EVALUATION ISN'T ENOUGH

- Model predicts board quality
- **But cannot choose moves effectively alone**
- Solution:
  - Add search algorithms:
    - **Minimax**
    - **Negamax**

Negamax



Minimax



NEGAMAX OPTIMIZATION

- Faster search via pruning & move ordering
- Reuse positions with caching (transposition table)
- Handle tactics with quiescence search
- **Negamax outperformed Minimax**

FINAL MODEL & CONCLUSIONS

- Final System
  - CNN evaluates positions
  - Negamax selects the best move
- Takeaways:
  - Evaluation alone does not strengthen gameplay
  - Search algorithms are essential
  - **Best results come from Neural Network + Search**

THANK YOU
QUESTIONS?